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Title: The Evolving ISM and Stellar Populations in Virgo Cluster Galaxies

Abstract: We propose a Treasury program to obtain NIRCam+MIRI imaging for 40 disk galaxies in the nearest massive galaxy cluster, Virgo. These galaxies have unparalleled complementary data from HST, ALMA, AstroSat, and VLT/MUSE, and they cover diverse morphology, orientation, and infall stages into the cluster. Our JWST program uses an efficient design to achieve several major science goals at once: (1) obtain recent star formation histories as a function of infall stages, resolving long-standing debates on the timescale of environmental quenching and star formation efficiency variations; (2) trace the rapidly evolving ISM at unprecedented depth and resolution, characterizing the internal and environmental processes that redistribute, remove, and destroy dust and gas; (3) determine precise relative distances to all targets, constraining their orbital histories in Virgo as well as the assembly of the Virgo Cluster itself. Together these will significantly advance our understanding of galaxy evolution in cluster environments. We will rapidly release reduced images and stellar cluster / HII region catalogs to support diverse, high-impact science from the community.

Scientific Category: Nearby Galaxies to Cosmic Noon

Alternate Category: Gas, Dust and the ISM

Science Keywords: Disk galaxies, Galaxy clusters, Galaxy environments, Galaxy evolution, Interstellar dust, Star formation, Stellar populations

Science / Parallel / Charged Time: currently at 55.7 h / 7.6 h / 156.3 h

Sample Information: [Google Spreadsheet](#)

■ Scientific Justification (required for all)

Galaxy evolution is governed by both internal and environmental processes [10, 22]. Environmental effects are most evident in dense regions such as rich galaxy clusters, where infalling satellite galaxies rapidly lose their gas content and subsequently quench at an accelerated pace [12, 23]. These processes operate from very early cosmic times [$z > 5$; 52] and have profoundly shaped galaxy populations throughout the cosmic history [32].

Over the past decades, statistical analyses and individual target studies of cluster satellites have identified several quenching mechanisms, including ram pressure stripping (RPS) acting on the galaxies from the outside as well as feedback from stars and AGNs operating from within [23]. But the exact physical consequences of these processes remain debated, and important questions are unanswered: **How does star formation activity wax and wane as cluster satellites go through various infall stages? How do external RPS and internal feedback processes reshape the ISM in concert? How do these sub-galactic scale changes contribute to a galaxy cluster’s global evolution?**

We propose a JWST Treasury program targeting 40 disk galaxies in the Virgo Cluster to make new breakthroughs on these questions. With complete censuses of star clusters and HII regions in every galaxy, we will reliably reconstruct their recent star formation history (SFH) with unprecedented time resolution, *settling a long-standing debate* about quenching timescales and bursty SFHs in galaxy cluster satellites. With high-resolution, sensitive tracers of the evolving ISM, we will explain star formation variations with ISM physics on small scales, reveal the gas leaving galaxies in outflows or stripped tails, and *for the first time*, link dust evolution to enrichment of the intracluster medium (ICM). We will also get precise distances for all targets, mapping out their 3D distribution and connecting their infall with Virgo’s global assembly, *a feat only possible with JWST in Virgo* [$d=16.2$ Mpc; 4]. All this science will be achieved with one integrated program imaging stellar and dust continua, recombination lines, and PAH emission altogether.

This program is uniquely timed to build on major new multiwavelength surveys — *four large programs* on HST, ALMA, AstroSat, and VLT/MUSE provide complementary measurements of the unobscured stellar populations, gas kinematics, and metallicity for the same sample of Virgo galaxies [16, 72]. Despite this excellent multiwavelength coverage, *only JWST* has the capabilities in the IR to resolve *all* star clusters and HII regions and trace gas and dust *within and around* galaxies, which are instrumental for connecting small-scale star formation and feedback physics to galaxy-scale quenching, and even to galaxy cluster-scale evolution. The 40 target galaxies span all infall stages into Virgo [inferred via phase-space information, gas content and morphology, and orbital modeling; 25, 76, 85] while marginalizing over galaxy-to-galaxy variations and orientation effects. A *stellar mass-matched field control sample* also exists (Fig. 1) with published JWST, HST, ALMA, and VLT data by other surveys [e.g., PHANGS; 27, 36, 37, 41], enabling rigorous distinction between environmental and internal processes across the full range of galaxy conditions.

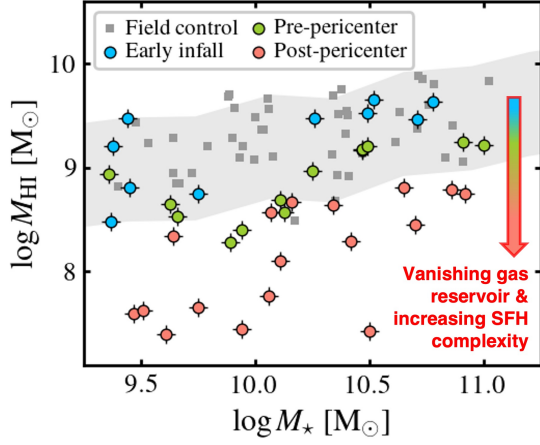


Fig. 1: We target 40 Virgo galaxies in different infall stages (color filled points). Those in more advanced infall stages retain less HI, reflecting environmental processing of their ISM. Existing JWST data for field galaxies (from PHANGS; [37, 39], dark gray symbols) with similar stellar masses and “normal” HI content (gray shaded area; [17]) provide an ideal control sample for separating internal versus environmental effects.

Goal #1. Definitive recent SFH with star clusters and IR SFR tracers. We will conduct a complete census of star clusters in Virgo galaxies with JWST NIRCам and new public HST data [72], nailing down their recent SFH with the cluster age distribution [73, see Fig. 2]. Only JWST can find *all* the youngest star clusters ($\lesssim 10$ Myr), which host much of the young stellar mass but remain embedded and often invisible in UV/optical [63]. JWST and HST photometry together can break the age–extinction degeneracy in SED fitting for star clusters [83], greatly improving the accuracy of mass and age determination in *all* age bins. With complete catalogs of star clusters and reliable age measurements, we will robustly constrain the recent SFH (up to ~ 100 Myr) and settle the debate on whether SFR is briefly enhanced by RPS within 10s of Myr around the first pericenter passage, and whether the quenching phase is rapid ($\lesssim 100$ Myr) or more prolonged [Fig. 2 right; 12, 23].

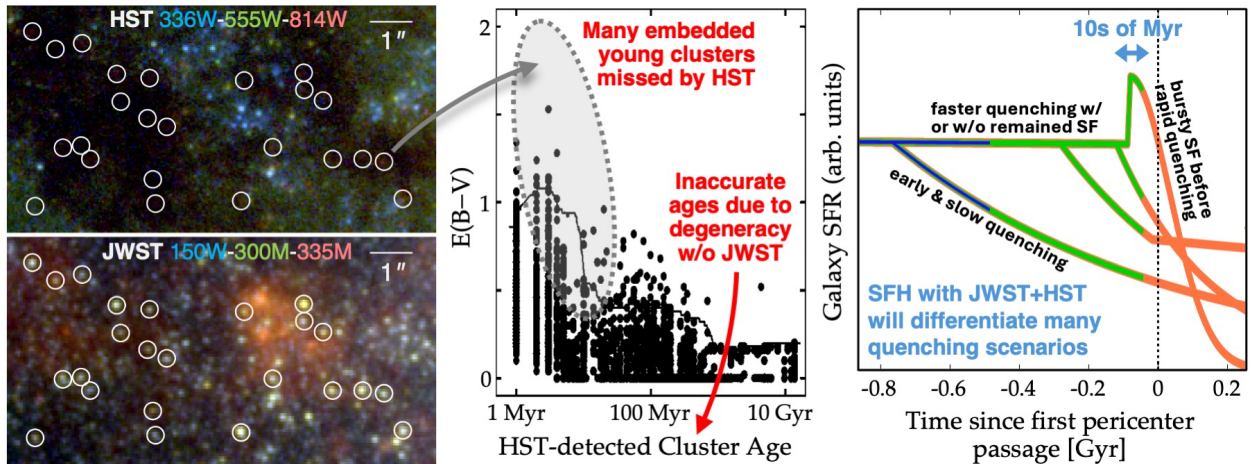


Fig. 2: (Left) HST and JWST images for a small region in a Virgo galaxy [36, 37]. Many embedded star clusters are only visible to JWST. (Middle) Age–extinction distribution of HST-detected star clusters in the same galaxy [73]. The high $E(B-V)$, young age region is severely incomplete without JWST [63]. HST-only age estimates are also biased due to the age–extinction degeneracy [82], limiting reliable reconstruction of SFH from the star cluster population. (Right) Example SFHs with various quenching scenarios [23]. JWST and HST together will better constrain SFHs across all age bins to differentiate these scenarios.

We will also derive recent SFR independently from JWST Pa α , Br α , and 21 μ m data, together with HST H α . Our measurements will resolve long-standing discrepancies in the inferred SFR and efficiency changes in Virgo satellites [14, 53, 75], which largely stem from the use of SFR tracers with different timescales and sensitivity to diffuse emission. JWST Pa α will *resolve* HII regions to 60 mas $\approx$ 5 pc scales and separate them from the diffuse ionized gas, a major contaminant for recombination line-based SFR [especially in ICM-affected systems; 77]. Combining Pa α , Br α , and HST H α enables complete, *local* accounting and correction for extinction, offering robust SFR on $\lesssim$ 10 Myr timescale. In addition, 21 μ m emission directly traces the dust-reprocessed emission and can be combined with new AstroSat UV data to derive SFR over longer, $\sim$ 100 Myr timescale [8]. These will replace previous estimates from GALEX UV and WISE mid-IR data [40], which suffer from much lower resolution and sensitivity; they will also be critical for interpreting the MUSE data, whose $\sim$ 1'' resolution still leaves many HII regions unresolved in H α , and whose stellar continuum primarily traces older stellar populations. With the high-resolution JWST SFR maps, we will further spatially correlate them with ISM structural properties (Goal #2) to find the *immediate cause* of the changing star formation rate across each galaxy [43].

Goal #2. Physical properties of the evolving ISM at high resolution. We will use JWST images of PAH emission at 3.3–11.3 μ m to trace the *redistribution, removal, and destruction of dust and gas* in Virgo satellites by internal and environmental processes. The mid-IR emission from stochastically heated PAH molecules is becoming a new workhorse for tracing the bulk neutral gas in the JWST era, as it is well mixed with both atomic and molecular gas [64] and can be calibrated via empirical and model-based methods [42, 45]. Its major advantage is exceptional spatial resolution, reaching 9–30 pc scales in Virgo, up to $10\times$ and $100\times$ *better* than the best CO and HI data [Fig. 3 left; 16, 21]. It is also sensitive down to $\lesssim$ 5 $M_{\odot}$ pc $^{-2}$ at such high resolution [64], unparalleled by other gas tracers.

The sharp, sensitive PAH maps enable a major step forward in quantifying ISM structure changes due to ICM–ISM interactions. This directly builds on low-resolution HI and CO studies capturing global leading/trailing side asymmetry [Fig. 3 top left; 60, 62] and HST extinction maps revealing substantial extra-planar dust in Virgo satellites [1, 24]. With JWST mid-IR maps, we can fully sample the small-scale gas column density PDF and contrast them with existing measurements for the field control sample [55]. This allows us to quantify the effects of gas compression (prevalence of high column density gas), stripping (lack of low column density gas), and even elevated turbulence (broadening of column density PDF) due to ICM–ISM mixing [20]. These small-scale changes in ISM properties are the immediate cause of SFR variations (Goal #1) and the key mediator of environmental processes, yet *no existing dataset provides full access to them* due to limited resolution and sensitivity. With our sample designed to cover satellites with diverse orientation angles and infall stages, we can *assess ICM–ISM interactions in all common configurations* and test simulation predictions given varying ICM wind strengths and impact angles [20, 35].

When low column density gas is stripped from cluster galaxies, stellar and AGN feedback are expected to launch gas outflows more easily. Our JWST program will put these predictions to the best test: MIRI PAH imaging will reveal neutral gas outflows in inclined

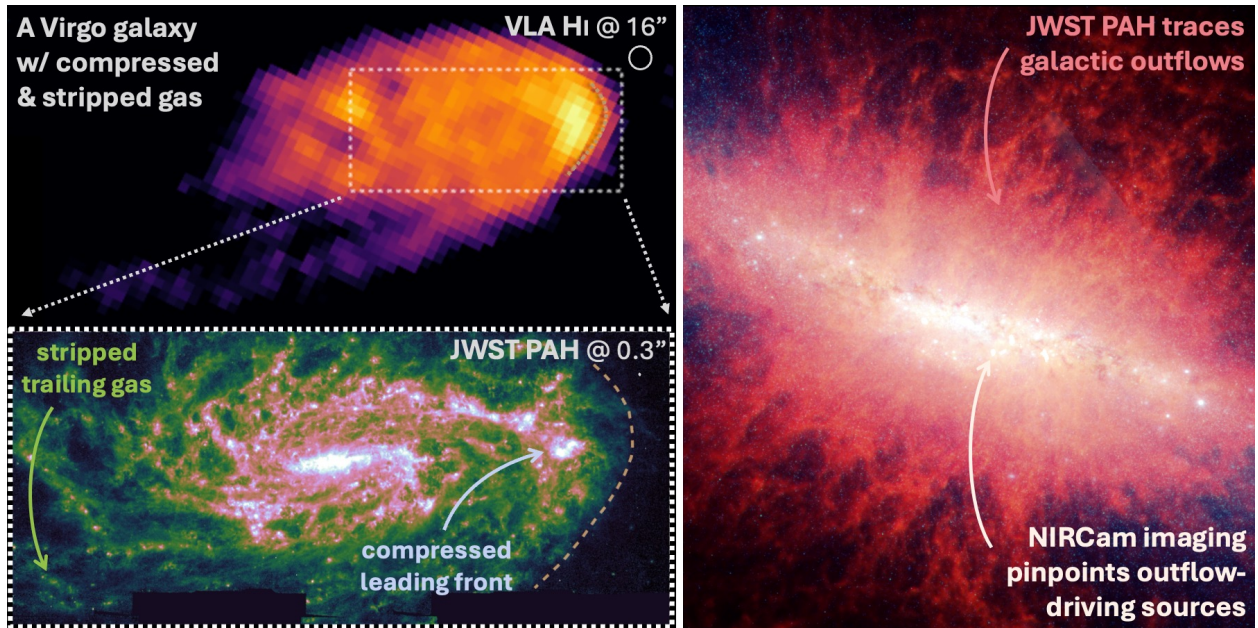


Fig. 3: (Top left) VLA HI image of a Virgo galaxy undergoing ram pressure stripping, with gas stripped to the East direction. (Bottom left) Existing MIRI data covering part of this galaxy [39], with $>50\times$ better resolution than HI. Bright PAH emission on the leading side marks spots of gas compression and clustered star formation, whereas diffuse PAH on the trailing side reveals stripped gas with little star formation. Our program will enable such anatomy for galaxies at all infall stages. (Right) PAH can also trace gas outflows [11], while NIRCcam imaging pinpoints their launching sites (central starbursts or AGN). Such outflow can be another key channel of gas removal (and thus quenching) in Virgo galaxies [71, 80].

systems [11, 71]. Meanwhile, with NIRCcam we will identify star clusters and HII regions and estimate their energy output [5, 55], thereby evaluating them as potential powering sources for the outflows (Fig. 3 right). This will provide a much-needed systematic view that expands on recent serendipitous discoveries of gas outflows in several Virgo satellites [71, 80].

As the ISM gets stripped or launched from a satellite galaxy, the dust abundance and composition change due to sputtering [51]. This ultimately sets the amount and composition of dust present in the ICM [47]. We will put direct constraints on the evolution of small grains as they join the ICM by estimating the PAH abundance, size distribution, and ionization from JWST mid-IR band ratios [18, 70]. Such constraints will shed light on PAH properties far outside the ISM regime, which will be widely useful for interpreting PAH emission around galaxies in the local universe, as well as those serendipitously detected in JWST campaigns targeting more distant cluster lensing fields [9, priv. comm.].

Goal #3. Precise distance measurements with TRGB. We will use deep NIRCcam imaging in F090W and F150W to pinpoint distances to all targets with the “tip of the red giant branch” (TRGB) method [38]. Recent successful JWST programs with similar setups have achieved 2–3% relative precision [3, 54]. This corresponds to 0.3–0.5 Mpc at Virgo’s distance [16.2 Mpc; 4], comfortably resolving the full depth of the cluster [$2R_{200} \approx 3$ Mpc; 28]. This represents a transformative advance over prior distance estimates from group

membership [34], Tully–Fisher relation [74], or surface brightness fluctuations [50], all of which have $\gtrsim 10$ –20% errors and thus little resolving power across Virgo. This huge leap in precision is critical for robustly estimating the local ICM density around each galaxy, which directly sets ICM ram pressure and thus the gas stripping rate [76]. Accurate 3D positions are also key to constraining galaxy orbits and infall stages in Virgo [68, 76], thereby anchoring their evolutionary timeline. *These contextual constraints are critical for modeling ISM evolution and galaxy quenching during cluster infall.*

Beyond individual galaxy’s evolution, precise distances and orbital reconstruction across our sample will shed light on the assembly of the Virgo Cluster itself. This is especially timely: JWST TRGB distances have just been obtained for several elliptical galaxies in Virgo [4] and will soon be measured for a larger set of galaxies in Virgo’s central substructure [81, J-Virgo]. The disk galaxies in our sample are instead newcomers spread over 100 deg^2 in Virgo, across and beyond its virial radius. With numerical action reconstruction [68, 69], we will constrain their inflow motions to complete a 3D view on Virgo’s dynamic architecture. This analysis is uniquely possible for Virgo thanks to its proximity, spatial expanse, and ongoing accretion. *Only JWST has the required resolution and sensitivity to accomplish this goal:* measuring TRGB at similar accuracy with HST would require upwards of 16 orbits per target for WFC3/UVIS [6, 30], making it prohibitive to carry out across 40 targets.

Why Treasury: Our program will yield a legacy dataset with broad science applications and impacts well beyond those discussed above. The rich filter combination, deep integration, and large, well-characterized sample will enable studies of ISM energetics [26], AGB stars [31], serendipitous supernovae detections [49], and identification of supernova progenitors or host systems of other transient events [57]. The program is also time critical – it is urgently needed for making best use of substantial ongoing investments by JWST, HST, AstroSat, ALMA, and VLT on Virgo, as well as for contextualizing multiple large programs on rare “jellyfish” galaxies in local clusters [56, 78, GTO 1178] and satellites in higher- z clusters [58, 79, GO 3224]. We believe treasury status is appropriate given these considerations.

■ Technical Justification (required for GO, DD and Survey only)

Observation Design: We propose an efficient imaging program with NIRCam and MIRI to support our three main science goals. The filter choices and imaging strategies have been proven effective by many programs in previous cycles [e.g., 3, 37, 39, 84].

We use NIRCam’s F090W, F150W, F300M, and F430M together with public HST broad-band data to capture the continuum emission of star clusters at all ages. The integration depths are sufficient to detect most $10^3 M_\odot$ clusters with $A_V \lesssim 5$ at $S/N > 10$ [based on CIGALE SED models; 37], or more massive / less embedded clusters at higher significance. We will also leverage medium- and narrow-band data tracing PAH and hydrogen recombination lines (F335M, F187N, F405N, see below) for distinguishing young clusters from older counterparts [33, 63, 73]. These setups directly support Goal #1.

We use NIRCam’s F335M and MIRI’s F770W & F1130W filters to trace PAH emission [18]. We leverage F300M to subtract continuum from F335M, and F1000W to subtract

continuum from F770W and F1130W. For the MIRI filters, we target $S/N = 10$ for extended emission at $1\text{--}2\text{ MJy/sr}$, which probe diffuse gas down to $\lesssim 5\text{ M}_\odot\text{ pc}^{-2}$ at $20\text{--}30\text{ pc}$ scales [64]. For NIRCam’s F335M, the integration results in $S/N = 6$ detection for similarly diffuse gas [65] *at a sharper $<10\text{ pc}$ resolution*. We also use MIRI’s F2100W filter to trace continuum emission by hot dust, which is key to accounting for embedded star formation, constraining dust-reprocessed radiation pressure in HII regions, and (when combined with PAH tracers) gauging PAH abundance variations [8, 19, 55]. We target $S/N \approx 10$ at 1 MJy/sr to detect emission from both star-forming regions and diffuse gas [following 37]. These designs ensure sensitivity to the evolving ISM within and around galaxies, directly supporting Goal #2.

We use NIRCam’s F187N ($\text{Pa}\alpha$) and F405N ($\text{Br}\alpha$) filters to detect and resolve individual HII regions, which will complete the inventory of *all* star formation, constrain local dust extinction, and pinpoint feedback and outflow launching sites [15, 44, 46]. We will combine F150W, F300M, and F430M to model and remove continuum in these narrow bands. The paired F187N+F405N integration will achieve $>10\sigma$ detection for faint HII regions down to 4×10^{-16} and $4\times 10^{-17}\text{ erg/s/cm}^2/\text{arcsec}^2$ in $\text{Pa}\alpha$ and $\text{Br}\alpha$; this depth is also necessary to overcome $1/f$ noise, a key concern for narrow-band imaging (priv. comm.). Note that for a fraction of our targets the $\text{Pa}\alpha$ and $\text{Br}\alpha$ lines would partly shift outside the narrow filter transmission curves due to large recession velocities. We drop the F187N+F405N integration for these targets and resort to HST F657N + MIRI F2100W for extinction corrected SFR.

We use NIRCam’s F090W and F150W filters to detect RGB stars and derive precise TRGB distances. F090W is optimal due to TRGB’s stability over a wide range of age and metallicity, and F150W allows for color selection at comparable resolution [3, 54]. Our integration scheme targets $S/N = 5$ at 1.5^{mag} below TRGB at the distance of Virgo’s center [16.2 Mpc ; 4]. This ensures robust TRGB detection with Sobel filter [30] and maximum likelihood [48] techniques throughout Virgo’s spatial span [$2R_{200} \approx 3\text{ Mpc}$; 28]. The TRGB apparent magnitude will be converted into distance based on the zero-point calibration obtained for the megamaser galaxy NGC 4258, for which a geometric distance is well measured [3, 59]. Our ETC uses setups recommended by [66] to get more realistic (though still slightly conservative) estimates for PSF photometry. We carefully position the two NIRCam modules so that at least one of them covers the outer galaxy halo [following 29, 61] for improved accuracy and avoiding issues related to crowding and young stellar populations [7].

Data Processing and Dissemination: We will aim for rapid data reduction over $1\text{--}2$ week timescale to enable community-engaged, time-sensitive science [e.g., serendipitous transient detections; 49]. Such fast reduction is made possible by community-built image processing tools [84], which we will adapt and improve in a public GitHub repository. We will then release finalized, quality-assured images and high-level data products (including star cluster / HII region catalogs and value-added measurements), accompanied by peer-review publications. We will plan our first releases around 6 months after project completion, and a final release including all high-level data products within 24 months. These products will be published online at CADC, with links to ancillary data from HST [72], VLT [16], ALMA [13], and VLA [21]. It will become a rich database with long-term accessibility and substantial legacy value, enabling continued science exploitation by the entire community.

■ Special Requirements (if any)

We constrain the position angle (PA) of the NIRCcam and MIRI fields for each target so that the galaxy disk can fit in the FoV at all allowed PA. All observations remain schedulable under these PA constraints. We also adjust the NIRCcam pointing coordinates to guarantee coverage of both the galaxy disk and part of the outer halo (for distance measurements).

We request the NIRCcam and MIRI observations for each target be taken uninterrupted so that the MIRI off-target observations (in parallel with NIRCcam) can be used for background removal for the MIRI on-target observations (see below).

■ Justify Coordinated Parallel Observations (if any, GO/DD only)

We require coordinated parallels to obtain MIRI off-target observations during NIRCcam on-target imaging. The off-target observations are needed because the (separate) MIRI on-target observations will often see emission across the entire field-of-view, leaving little signal-free area for background determination [see 84]. The NIRCcam-on/MIRI-off parallel imaging strategy would allow us to most efficiently achieve our science goals.

■ Justify Duplications (if any, GO/DD only)

We checked the JWST archive for existing imaging data that fully cover our targets and reach sufficient depth in any of the NIRCcam and MIRI filters. We found such data for 12/40 of our targets (mostly with MIRI). In such situation, we remove the duplicated filters (or entire instrument) from our request and will use the archival data instead.

We note several cases where archival data exist but do not meet our requirements: (1) F150W from GO 7763 [J-Virgo; 81] for eight targets: These observations reach adequate depth for TRGB distances (Goal #3), but they prioritize galaxy halo coverage at the expense of full inner disk coverage. The latter is essential for complete star cluster census for our science (Goal #1). (2) F150W and F187N from GO 3707 [PHANGS; 39] for eight targets: The F150W data neither cover the galaxy halo nor have adequate depth for reliable TRGB distances. The F187N data are $\sim 4\times$ shallower than needed for suppressing $1/f$ noise and obtaining reliable, continuum-subtracted $\text{Pa}\alpha$ measurements (for Goal #2). (3) F300M and F335M from GO 2107 [37], 3707 [39], and 4793 [67]: These data reach reasonable (slightly shallower) depth but do not fully cover our targeted footprint for complete star cluster census (Goal #1). These two filters are also the LW counterparts of F090W in our program and therefore not driving the integration time requirements. (4) HST/ACS F814W parallel imaging from GO 18103 [72]. With their primary WFC3 pointings centered on the galaxy disks, the vast majority of parallel ACS pointings fall far outside the halo of these galaxies. In the few cases of galaxies with larger angular sizes, the ACS pointing covers a portion of the halo with relatively shallow depth, which would allow for TRGB measurements with at best 10% precision [2], insufficient for our science (Goal #3). In the above cases, we keep our design unchanged and will fold in archival data in our analyses when possible.

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